DIFFERENTIAL DISTILLATION

Name: Purva Kaushalbhai Shah

Group members name:
Gaurav Budhwani
Rachana Durgam

Table of Contents:

Objective:
The aim of this experiment is to separate a binary mixture of water and acetic acid using simple distillation and determine the moles of acetic acid through titration and to verify the Rayleigh equation for simple distillation in a binary system.

Theory and Background:
Distillation is the process of separating two or more components from a liquid mixture by the process of vaporization and condensation. It is a widely used separation technique to separate two or more liquids having significant differences in their boiling points. The fundamental concept behind distillation is that when a liquid mixture is heated, the component with the lower boiling point evaporates first. These vapors are then condensed and collected separately, enabling the purification of the mixture.

In the case of simple distillation, also known as differential distillation, the vapor is immediately directed into a condenser, where it cools and turns back into the liquid phase (distillate). The resulting distillate is not completely pure but retains the same composition as the vapor at a given temperature and pressure. This follows Raoult's Law, which states that the partial vapor pressure of each component in a liquid mixture is proportional to its mole fraction and its vapor pressure in the pure state.

Simple distillation works best when there is a significant difference in the boiling points of the components, usually around 25°C or more. It is mainly used to separate volatile liquids from non-volatile solids or oils. However, when the boiling points of the components are too close, more specialized methods like fractional distillation, azeotropic distillation, or vacuum distillation are needed for effective separation.

To describe the relationship between the composition of the residue liquid, the distillate, and the volatile components during the distillation process, Rayleigh's Equation is used. This equation is derived from the mass balance principle and the concept of vapor-liquid equilibrium (VLE).

Rayleigh Equation can mathematically be written as: $\ln\left(\frac{F}{W}\right) = \int_{Xw}^{XF} \frac{dx}{y^* - x}$

Where F $is\ total\ moles\ in\ the\ feed$, $D\ is\ the\ total\ moles\ in\ the\ distillate$,
$W\ is\ the\ total\ moles\ in\ residue$, $x_F\ is\ the\ mole\ fraction\ of\ water\ in\ the\ feed$,
$x_W\ is\ the\ mole\ fraction\ of\ water\ in\ the\ residue$

Derivation:
Mole balance: $In + Out - Generation = Accumulation$
$Moles\ in = 0$

$Moles\ out\ =\ dD$
$Moles\ accumulated\ =\ dL$
$Moles\ generated\ =\ 0$
$\Rightarrow\ dD\ =\ -\ dL$ (1)

For more volatile component: $y\ *\ dD\ =\ -\ d\,(Lx)\ =\ -\ Ldx\ -\ xdL$ (2)
From (1) and (2) $-\ y\ *\ dL\ =\ -\ Ldx\ -\ xdL$
$\Rightarrow\ -\ Ldx\ =\ xdL\ -\ y\ *\ dL$
$\Rightarrow\ -\ Ldx\ =\ (x\ -\ y\ *)dL$
$\Rightarrow \frac{dL}{L}\ =\ -\frac{dx}{x-y^*}$

On integrating, we get : $\int_{W}^{F}\frac{dL}{L}\ =\ \int_{x_w}^{x_F}\frac{dx}{x-y^*}\ \Rightarrow\ ln\,(\frac{F}{W})\ =\ \int_{x_w}^{x_F}\frac{dx}{y^*-x}$

Where $F\ is\ total\ moles\ in\ the\ feed$, $D\ is\ the\ total\ moles\ in\ the\ distillate$, $W\ is\ the\ total\ moles\ in\ residue$, $x_F\ is\ the\ mole\ fraction\ of\ water\ in\ the\ feed$, $x_W\ is\ the\ mole\ fraction\ of\ water\ in\ the\ residue$

Overall balance: $F\ =\ D\ +\ W$
Mole balance on water: $Fx_F\ =\ Dx_D\ +\ Wx_w$

<u>Apparatus:</u>

1. Electrical heating system
2. Thermometer
3. Conical flask
4. Pipette
5. Burette
6. Acetic acid.
7. Simple distillation unit without reboiler
8. Distilled water

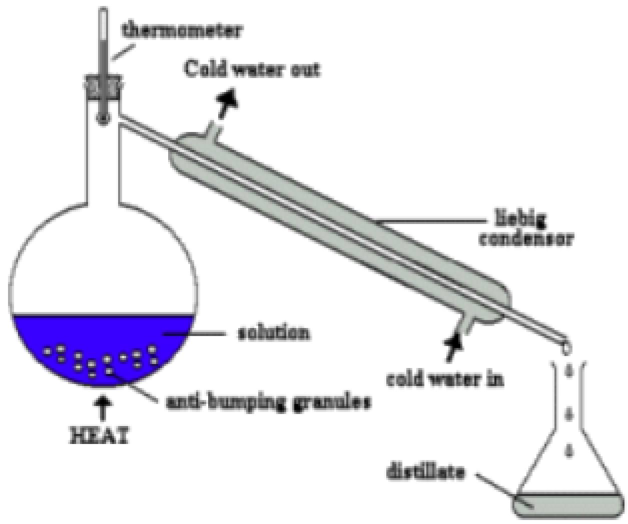

Figure 0: Schematic diagram of the experimental apparatus

Procedure:

1. Prepare a 500 mL feed mixture by combining acetic acid (300 mL) and water (200 mL).
2. Transfer the mixture into the distillation flask.
3. Seal the flask with a thermowell and insert a thermometer.
4. Begin the flow of cooling water through the condenser and gradually heat the mixture. Once boiling starts, adjust the heating rate to ensure distillate collection at approximately 30 drops per minute, gathering a total of 50 mL per sample.
5. Collect at least 5 distillate samples.
6. The initial samples will have a higher concentration of the more volatile component, which will decrease in subsequent fractions.
7. Wrap the samples in aluminum foil to prevent evaporation due to high temperatures.
8. Titrate each sample using a 1N NaOH solution.
9. Determine the mole fractions of acetic acid and water from the titration results.
10. Calculate the equilibrium data y* corresponding to x_w.
11. Plot $y^*\ vs\ x_w$ curve.
12. Plot $\frac{1}{(y^* - x)}\ vs\ x$
13. Find area under curve for $\frac{1}{(y^* - x)}\ vs\ x_w$.

Known parameters:

- $Volume\ of\ acetic\ acid\ =\ 300\ mL$
- $Volume\ of\ water\ =\ 200\ mL$
- $Density\ of\ acetic\ acid\ =\ 1.05\ g/mL$
- $Density\ of\ water\ =\ 1\ g/mL$
- $Normality\ of\ NaOH\ =\ 1\ Normal$
- $The\ boiling\ point\ of\ water\ =\ 100ºC$
- $The\ boiling\ point\ of\ acetic\ acid\ =\ 117.9\ ºC$

From the boiling points, we can say that here, the more volatile component is water.

Formulae:

- $N_1V_1\ =\ N_2V_2$
  - Where N_1 $is\ the\ Normality\ of\ NaOH$
  - V_1 $is\ the\ Volume\ of\ NaOH$
  - V_2 $is\ Volume\ of\ Distillate$
  - $N_2\ =\ \frac{N_1V_1}{V_2}\ =\ Normality\ of\ acetic\ acid\ in\ distillate$
- $F\ =\ D\ +\ W$
  - Where $F\ is\ total\ moles\ in\ the\ feed$
  - $D\ is\ the\ total\ moles\ in\ the\ distillate$
  - $W\ is\ the\ total\ moles\ in\ residue$
- $Fx_F\ =\ Dx_D\ +\ Wx_w$
  - Where x_F $is\ the\ mole\ fraction\ of\ water\ in\ the\ feed$
  - x_D $is\ the\ mole\ fraction\ of\ water\ in\ the\ distillate$
  - x_W $is\ the\ mole\ fraction\ of\ water\ in\ the\ residue$
- $\ln\left(\frac{F}{W}\right)\ =\ \int_{x_W}^{x_F}\frac{dx}{y^*-x}$
  - Where $F\ is\ total\ moles\ in\ the\ feed$
  - $D\ is\ the\ total\ moles\ in\ the\ distillate$
  - $W\ is\ the\ total\ moles\ in\ residue$
  - x_F $is\ the\ mole\ fraction\ of\ water\ in\ the\ feed$
  - x_W $is\ the\ mole\ fraction\ of\ water\ in\ the\ residue$
  - $y\ *\ =\ composition\ of\ water\ in\ vapour\ phase$

- $x = composition\ of\ water\ in\ liquid\ phase$

Readings:

TABLE 1: TABLE CONTAINING EXPERIMENTAL DATA

Normality of NaOH (N1)	Volume of NaOH solution (V1) ml	Volume of Acetic Acid (V2) ml	Normality of Acetic Acid (N2)
1	12.67	2	6.335
1	13.32	2	6.66
1	14.3	2	7.15
1	15.35	2	7.675
1	16	2	8

TABLE 2: TABLE CONTAINING CALCULATED DATA OF DISTILLATE

Moles of Acetic Acid	Moles of water	Total moles	Y (acetic acid)	Y (water)
0.31675	1.800154321	2.116904321	0.1496288693	0.8503711307
0.333	1.75	2.083	0.1598655785	0.8401344215
0.3575	1.674382716	2.031882716	0.1759451947	0.8240548053
0.38375	1.593364198	1.977114198	0.1940960216	0.8059039784
0.4	1.543209877	1.943209877	0.2058449809	0.7941550191

TABLE 3: TABLE CONTAINING CALCULATED DATA OF RESIDUE

W(moles)	xaa	xw
18.04976235	0.1763596627	0.8236403373
18.08366667	0.1751304123	0.8248695877
18.13478395	0.17328577	0.82671423
18.18955247	0.1713208725	0.8286791275
18.22345679	0.1701104261	0.8298895739

TABLE 4: TABLE CONTAINING VLE DATA FOR ACETIC ACID AND WATER

y*	x
0	0.001
0.253	0.01

0.425	0.02
0.624	0.05
0.755	0.1
0.798	0.15
0.815	0.2
0.83	0.3
0.839	0.4
0.849	0.5
0.859	0.6
0.874	0.7
0.898	0.8
0.935	0.9
0.963	0.95
1	1

TABLE 5: TABLE CONTAINING CALCULATIONS USING EQUILIBRIUM CURVE

y*	y*-x	1/(y*-x)	ln(F/W)	Integral
0.9067469248	0.08310658753	12.03274048	0.1108985581	0.1491642282
0.9072017474	0.08233215978	12.14592211	0.1090219396	0.08397578227
0.9078842651	0.08117003508	12.31981727	0.1061992171	0.01431281158
0.9086112772	0.0799321497	12.51061061	0.1031836873	0.119653463
0.9090591423	0.07916956846	12.63111596	0.1013214775	0.1848938679

Calculations:
For table 1:
$N_1V_1 = N_2V_2$

- Where N_1 $is\ the\ Normality\ of\ NaOH$
- V_1 $is\ the\ Volume\ of\ NaOH$
- V_2 $is\ Volume\ of\ Distillate$
- $N_2 = \frac{N_1V_1}{V_2} = Normality\ of\ acetic\ acid\ in\ distillate$

$N_2 = \frac{N_1 V_1}{V_2} = \frac{12.67 * 1}{2} = 6.335$

For table 2 and 3:
$Density\ of\ acetic\ acid = 1.05\ g/mL$
$Density\ of\ water = 1\ g/mL$

Feed contains 300 mL of acetic acid and 200 mL of water.
Moles of acetic acid in feed = (300*1.05)/60 =3.5
Moles of distilled water in feed = (200*1.00)/18 =16.66666667
Total moles in feed = 4.95+12.5 = 20.16666667

Next we calculated the moles by using normality, volume and knowing the n (valency, charge they give -> typically the factor which divides molarity to give normality).

Then we plotted the equilibrium curve by data provided on the internet.
And found the relationship between our x_w and y *.

After calculating y *, x_F and x_w and putting it as a limit for integration of $\frac{dx}{y^* - x}$ using the equation we obtained by plotting from Figure 2, the graph of $\frac{1}{y^* - x}$ vs x, we can get the value of $\int_{x_W}^{x_F} \frac{dx}{y^* - x}$.

We know F and W, $ln\left(\frac{F}{W}\right) = 0.1061249759\ for \int_{x_W}^{x_F} \frac{dx}{y^* - x} = 0.1104000306$

Error = $\frac{ln\ (F/W) - \int_{Xw}^{Xf} dx/(y^* - x)}{ln\ (F/W)}$ =0.00403
Error (%) = |0.00403| * 100 = 4.03 %

Results and discussion:

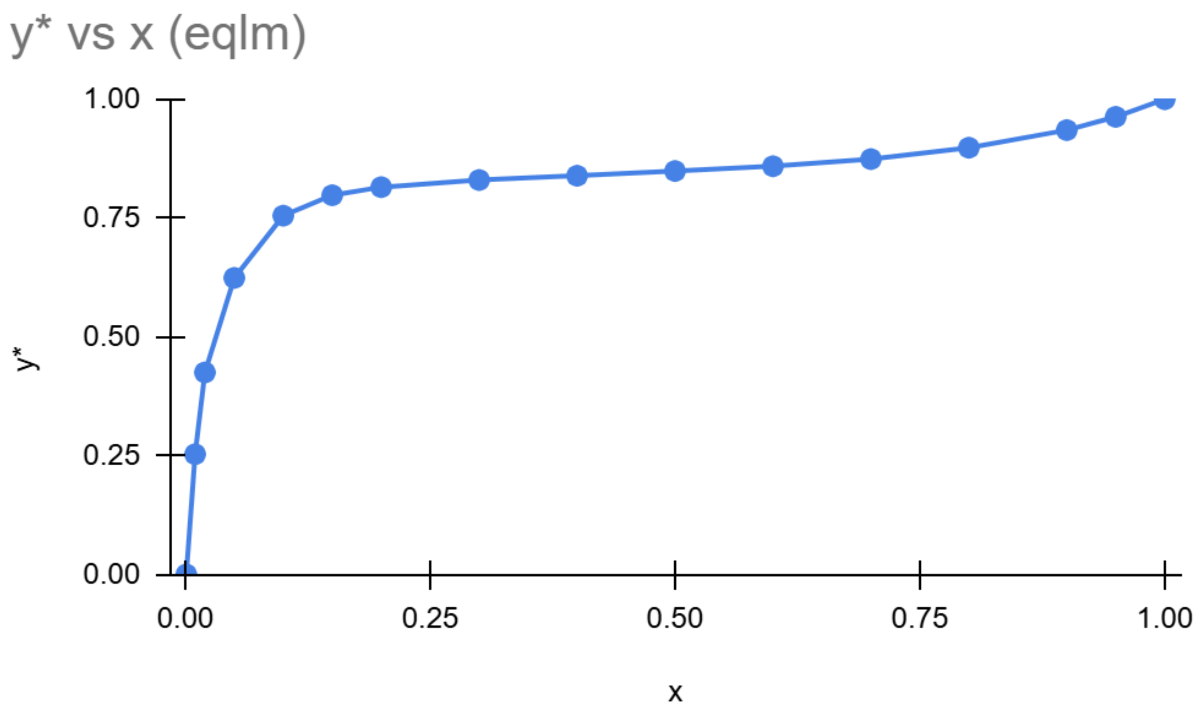

Figure 1: Figure showing the VLE equilibrium curve

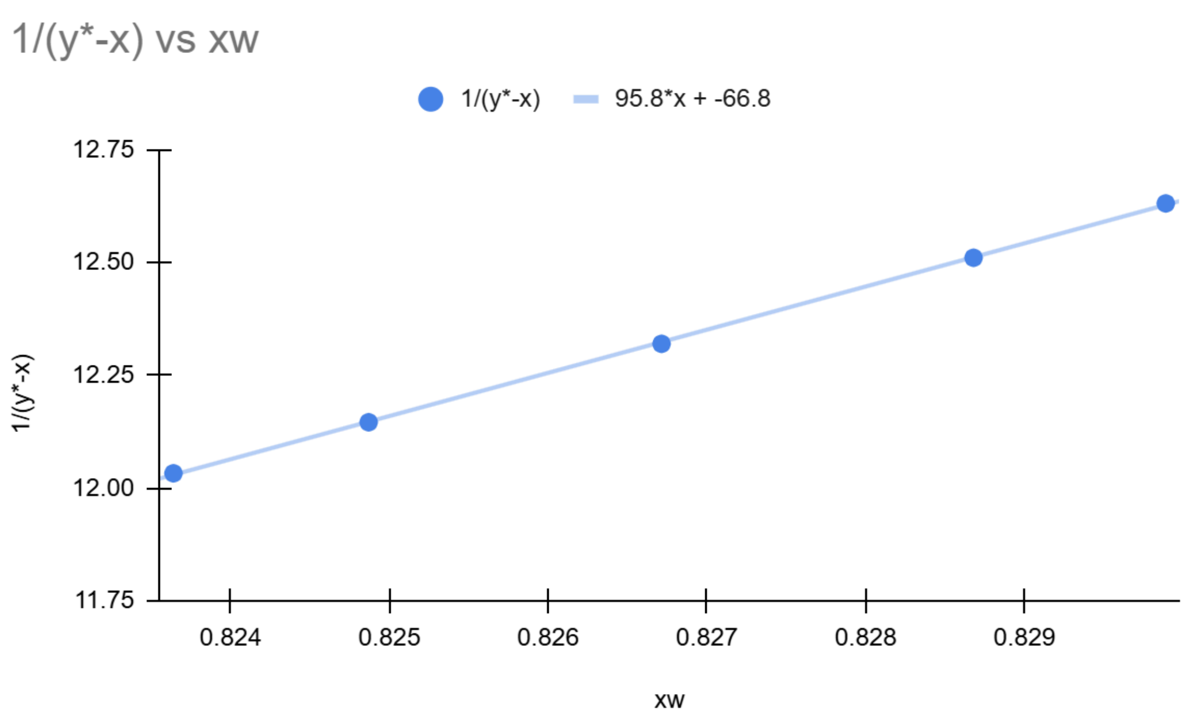

Figure 2: Figure showing the plot between $\frac{1}{y^* - x}\ vs\ x_w$

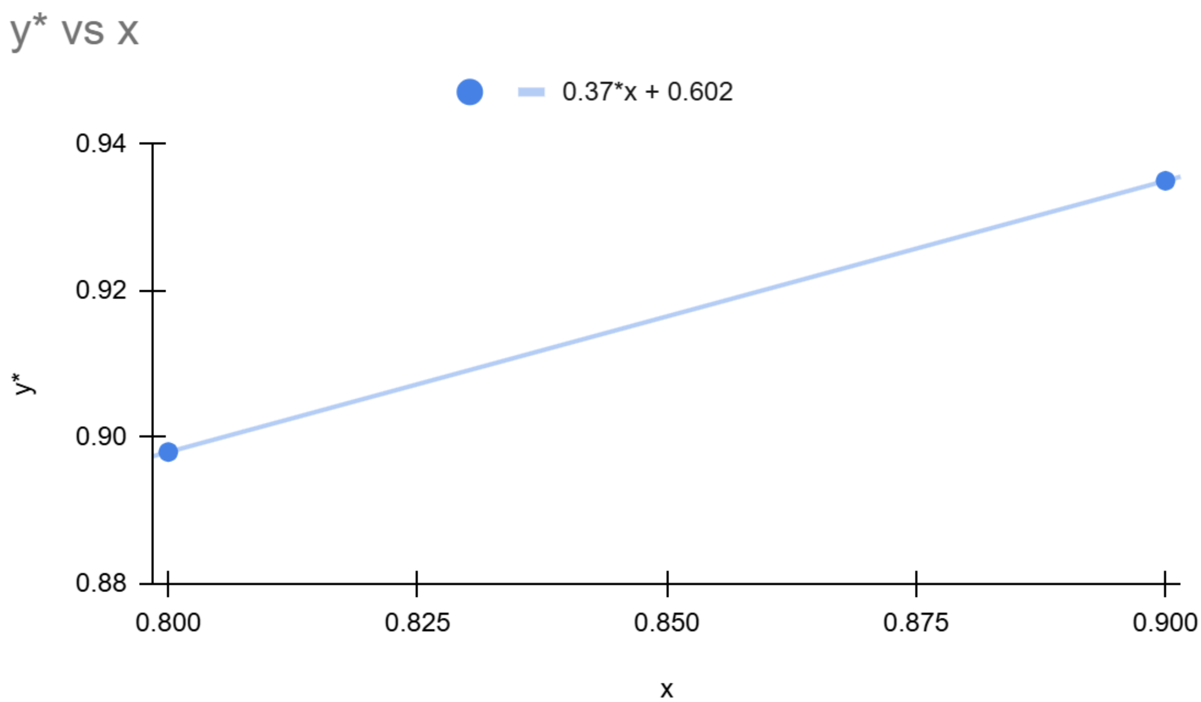

Figure 3: Figure showing the plot between y vs x*

Initially, the distillate contained a higher fraction of the more volatile component, water, as it evaporates first due to its lower boiling point. As distillation progressed, the concentration of acetic acid in the distillate gradually increased, corresponding to the depletion of water in the distillation flask. This trend was confirmed by the titration results, where increasing volumes of NaOH solution were required to neutralize the later distillate samples, indicating a higher acetic acid concentration over time.

To ensure consistency and accuracy, distillate samples were collected at regular intervals, with a fixed amount used for titration. Some discrepancies in results may have occurred due to evaporation losses, minor fluctuations in cooling water flow, or handling errors. However, the data closely followed the predictions from Rayleigh's equation, reinforcing the theoretical understanding of how the distillate and residue compositions evolve over time.

The gradual decrease in water content in later samples confirmed the continuous removal of the more volatile component, leading to a higher proportion of acetic acid in the remaining mixture. This behavior aligns with vapor-liquid equilibrium (VLE) principles and demonstrates the effectiveness of simple distillation in separating binary mixtures based on volatility differences. The low error margin of 4.03% further validates the accuracy of the experiment.

Conclusion:

The experiment successfully validated Rayleigh's equation for simple distillation and demonstrated the time-dependent variation in distillate composition. In the initial samples, the more volatile component, water, was present in higher concentrations, as indicated by lower NaOH consumption during titration. As distillation progressed, later samples were collected at higher temperatures, leading to an increased concentration of acetic acid in the distillate, which required more NaOH for neutralization. The results confirmed a gradual depletion of the more volatile component in the distillation flask, aligning with the expected trend.

Simple distillation proved to be an effective method for separating binary mixtures, reinforcing its practical applications in industrial and laboratory settings. The calculated values of $\int_{x_W}^{x_F} \frac{dx}{y^* - x}$ and $\ln\left(\frac{F}{W}\right)$ showed an error of 4.03%, which is significantly lower than the acceptable range of 15-30%, indicating high accuracy in the experiment.

Sources of errors:

- We may have taken more than 2mL samples.
- We may have not cleaned the apparatus properly.
- We may have not noticed the slightest pink colour during the titration and could have added more NaOH.
- The rate of cooling water was fluctuating.

References:

[1]"Wolfram Demonstrations Project," *Wolfram.com*, 2025. https://demonstrations.wolfram.com/VaporLiquidDataForAceticAcidWaterMixtureEffectOfGasPhaseDime/

[2] 'Lab Manual' provided in the classroom.

[3]BYJU'S, "Distillation," *Byjus.com*, 2015. https://byjus.com/chemistry/distillation/

[4]"rayleigh eqn distillation - Google Search," *Google.com*, 2024. https://www.google.com/search?q=rayleigh+eqn+distillation&rlz=1C1CHZN_enIN1037IN1037&oq=rayleigh+eqn+distillation&gs_lcrp=EgZjaHJvbWUyBggAEEUYOdIBCDUyMDhqMGo0qAIAsAIA&sourceid=chrome&ie=UTF-8